LX Foundry

stint

Track: Best Use of x402 on Solana

Solana Hackathon 2026 · Encode Club

THE GAP

Agents also rent time on a connection

A headless browser session, today's most common case

A live data stream

One pass, paid up front for the whole connection



the agent leaves early nothing comes back

Today, that time is often a pass paid up front. If the agent leaves early, nothing comes back.

THE DEMAND IN X402

x402 builders have asked for metered sessions with a refund

x402 issue #2273

metered-session

Prepaid metered sessions with auto-refund for streaming x402 transports.

Extension proposal, opened 2026-05-12

x402 PR #1231 · x402 over WebSocket

“defeats the reason to use WS”

A comment on the WebSocket transport, on a payment for every frame.

402 402 402 402

x402-foundation/x402 on GitHub, as of 2026-11-04

Stint: rent time on live connections, and get back what you don’t use.

An AI agent pays once over x402 to open a time-boxed session on a live service. A Solana program holds the USDC, pays the seller for the time used, and refunds the rest on-chain when the agent closes early.

THE FLOW

Three changes to the x402 flow



On the 402

A scheme for time

The offer carries a price per second, the time bought, a minimum and a maximum.

`scheme: "elapsed"`

On payment

One transaction

The agent's USDC moves into the session's vault and the session opens, together. Kora pays the fee, so the agent holds only USDC.

`fee payer: Kora`

At the connection

A gate

It checks the session on Solana, then opens the WebSocket. When the paid time ends, it drops the connection.

`close code: 4402`

FAILED PAYMENTS

A bad payment is refused with a reason

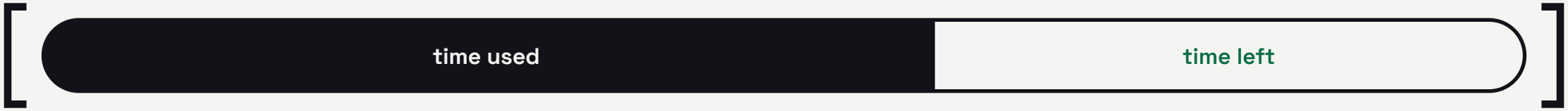
WHAT ARRIVES	WHAT HAPPENS	SCENARIO
Wrong terms, a wrong amount, or an extra instruction	A 402 with the reason, and no session on-chain `invalid_terms · invalid_amount · unexpected_instruction`	5
An accepted field one byte off the offer, or an old x402 version	Refused with the reason, and no session `invalid_payment_requirements · invalid_x402_version`	13
A payment that fails	A 402, and no session is created	6
The same payment, sent twice	One session on-chain, charged once; both answered with it	14

Each case runs end to end, on a local network and on devnet

CLOSING EARLY

The program refunds what is not used

One transaction, enforced by the Solana program



SELLER

Paid for the time used

At least the minimum the seller set.

AGENT

Refunded the rest

Back to the account the deposit came from.

Related work: ws402 (MIT, 2025) does prepaid WebSocket sessions over x402, with a refund sent by the seller's backend. In Stint a Solana program enforces the refund: the seller cannot keep it.

THE REST OF A SESSION

A session can end, extend and recover

The time runs out

The seller is paid in full. The gate stops forwarding and drops the connection with 4402.

scenario 3

A top-up

Before the end, the agent can add time to a running session, which moves the end. After the end, it is refused.

scenario 4

The gate is down

The client closes through the facilitator instead, or on-chain if it holds SOL.

scenarios 15, 24, 25

Nobody is needed to finish a session: once its paid time has ended, anyone may settle it.

THE LIVE DEMO

A live demo in Claude Desktop

Against a seller on devnet

1. Download the extension from the latest release.
2. Ask Claude to start the Stint demo.
3. Fund the agent's wallet with test USDC from Circle's faucet.
4. Launch a session, top it up, and close it early.

THE PANEL IN THE CHAT SHOWS

- The 402 exchange
- The paid time draining
- The wallet's log, with the refund
- Each step, linked to devnet's Explorer

The agent's wallet holds USDC only. Its spending limits are enforced in code.

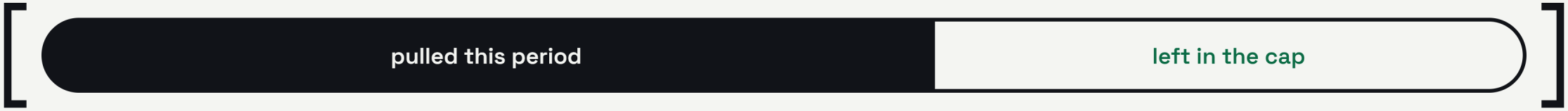
stint-demo.lxfoundry.ai sells a live stream of Solana account changes over WebSocket, up to 10 accounts at a time, at 0.001 USDC a second.

A real run on devnet, 2026-11-05: 44 of 200 s used, paid 0.044 USDC, refunded 0.156 USDC; 2 stream messages.

THE ALLOWANCE

A budget the agent cannot raise

The owner's allowance: a cap per period



1 Granted once

The owner sets a cap per period with the Solana Foundation's Subscriptions & Allowances program.

2 Pulled from it

Each open and top-up is pulled in the same transaction. An early close refunds the owner.

3 Refused past it

The program refuses the pull, so no session opens. The agent cannot raise its own budget.

WHAT YOU CAN REUSE

Every part is MIT, in one repository

`programs/stint`
The Solana program: open, top_up, claim, close. On devnet.

`@lxfoundry/stint-x402`
The elapsed scheme: offers, the payment, session tokens.

`@lxfoundry/stint-facilitator`
Verifies, settles through Kora, sweeps expired sessions.

`@lxfoundry/stint-gate`
Middleware in front of any long-lived connection.

`@lxfoundry/stint-client`
Spending limits enforced in code, whatever a model asks.

`@lxfoundry/stint-mcp`
The session tools, for any MCP host.

`allowance funding`
A command for owners to grant, revoke and read an allowance.

`@lxfoundry/stint-program-client`
Generated from the program's interface with Codama.

`examples/`
A seller, the Claude Desktop extension, an agent, and the stream.

Public repository: github.com/lxfoundry/stint

STATUS

Working end to end on devnet

BUILT AND CHECKED

`5iCYPqLjCWw5KzQTU6NBUhR2hxg5SWGmS1ctUn3qESvr`

The program is deployed on devnet, built verifiably, and its deployed bytes are checked against that build.

Payments, refusals, early closes, expiry, top-ups, fallback closes and allowance funding pass end to end, on a local network and on devnet.

BEFORE A PRODUCTION RELEASE

- Several gate instances
- Gate and agent restarts mid-session
- A settlement reply lost in transit
- Closes on an outage or admin command
- Closed or frozen payout accounts
- Mainnet hardening and limits

Each one is a GitHub issue, in the milestone “Production release”.

How it was built: one builder, with Claude Code. The system was specified before any code, then built in stages, each with a written plan, tests written before the code, and a review by agents that did not write it.

[▬D] stint

Stint: rent time on live connections, and get back what you don’t use.

CODE	github.com/lxfoundry/stint
LIVE DEMO	stint-demo.lxfoundry.ai
EXTENSION	github.com/lxfoundry/stint/releases/latest
DEMO VIDEO	youtu.be/PBlbQeD2K1M